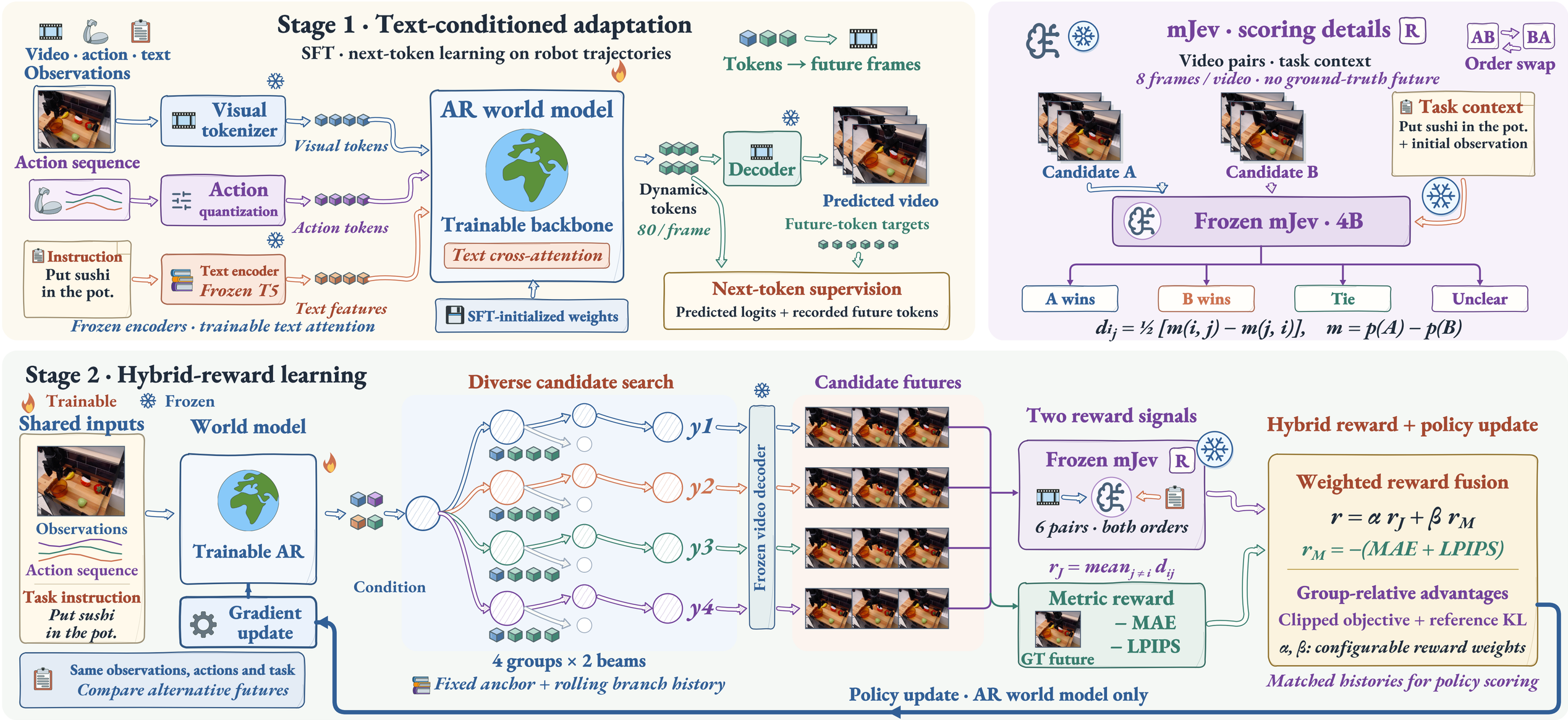

Stage 1 · Text-conditioned adaptation
SFT · next-token learning on robot trajectories
Video · action · text Observations
Action sequence
Visual tokenizer
Visual tokens
Action quantization
Action tokens
Instruction
Put sushi in the pot.
Text encoder
Frozen T5
Text features
Frozen encoders · trainable text attention
AR world model
Trainable backbone
Text cross-attention
SFT-initialized weights
Tokens → future frames
Dynamics tokens
80 / frame
Decoder
Predicted video
Future-token targets
Next-token supervision
Predicted logits + recorded future tokens
mJev · scoring details
R
Video pairs · task context
8 frames / video · no ground-truth future
AB
BA
Order swap
Candidate A
Candidate B
Task context
Put sushi in the pot.
+ initial observation
Frozen mJev · 4B
A wins
B wins
Tie
Unclear
$d_{ij} = ½ [m(i, j) − m(j, i)], \quad m = p(A) − p(B)$
Stage 2 · Hybrid-reward learning
Trainable
Frozen
Shared inputs
Observations
Action sequence
Task instruction
Put sushi in the pot.
World model
Trainable AR
Gradient update
Same observations, actions and task
Compare alternative futures
Diverse candidate search
Condition
y1
y2
y3
y4
4 groups × 2 beams
Fixed anchor + rolling branch history
Frozen video decoder
Candidate futures
Two reward signals
Frozen mJev
R
6 pairs · both orders
$r_J = mean_{j \neq i} d_{ij}$
Metric reward
GT future
– MAE
– LPIPS
Hybrid reward + policy update
Weighted reward fusion
$r = \alpha r_J + \beta r_M$
$r_M = -(MAE + LPIPS)$
Group-relative advantages
Clipped objective + reference KL
α, β: configurable reward weights
Matched histories for policy scoring
Policy update · AR world model only